higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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This marking guideline consists of 11 pages.

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QUESTION 1

1.1 1.1.1

$$y = \lim_{x\to 0} \frac{e^{-x^2} - \ln(1+x) - 1}{\cos x - \sin x - 1} \quad \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{2xe^{-x^2} - \frac{1}{1+x}}{-\sin x - \cos x} \checkmark$$

$$= 1 \checkmark \qquad (2)$$

1.1.2

$$y = \lim_{x\to 0} \frac{2^{-x} - 1}{3^{-x} - 1}$$

$$= \lim_{x\to 0} \frac{-2^{-x}\ln 2}{-3^{-x}\ln 2} \checkmark$$

$$= 0{,}631 \checkmark \qquad (2)$$

1.2

$$y = \frac{x^2 - x + 4}{\ln(x)}$$

$lnx \neq 0$

$x \neq 1 \checkmark$

$x \in R; x \neq 1 \quad or \quad (-\infty; 1) \cup (1; \infty) \quad or \quad \{x/x < 1 \cup x > 1\} \checkmark$ (2)

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QUESTION 2

2.1 $$f(x) = \frac{5}{3x+2}$$

$$f'(x) = \lim_{h\to 0} \frac{\frac{5}{3(x+h)+2} - \frac{5}{3x+2}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{5(3x+2)-5[3(x+h)+2]}{[3(x+h)+2](3x+2)}}{h} \checkmark$$

$$= \lim_{h\to 0} \frac{\frac{15x+10-15x-15h-10}{[3(x+h)+2](3x+2)} \checkmark}{h}$$

$$= \lim_{h\to 0} \frac{-15h}{[3(x+h)+2](3x+2)} \times \frac{1}{h} \checkmark$$

$$= \lim_{h\to 0} \frac{-15}{[3(x+h)+2](3x+2)}$$

$$= -\frac{15}{(3x+2)^2} \checkmark$$ (4)

2.2 2.2.1

$$y = \sin^5[\tan(\cos x^3)]$$

$$\frac{dy}{dx} = 5\sin^4[\tan(\cos x^3)] \checkmark \times \cos[\tan(\cos x^3)] \checkmark$$
$$\times \sec^2(\cos x^3) \checkmark \times (-\sin x^3) \checkmark \times 3x^2 \checkmark$$ (5)

2.2.2

$$y = \sqrt{x^3 + \sqrt{x^2 + \sqrt{3-x}}}$$

$$\frac{dy}{dx} = \frac{1}{2}\left[x^3 + \sqrt{x^2 + \sqrt{3-x}}\right]^{-\frac{1}{2}} \checkmark \times \left[3x^2 + \frac{1}{2}\left(x^2 + \sqrt{3-x}\right)^{-\frac{1}{2}}\right] \checkmark$$
$$\times \left[2x + \frac{1}{2}(3-x)^{-\frac{1}{2}}\right] \checkmark \times -1 \checkmark$$ (4)

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2.3 $$y = \frac{(9-3x)^4}{x^2\cos 2x}$$

$$\ln y = 4\ln(9-3x) - 2\ln x - \ln(\cos 2x)$$

$$\frac{1}{y}\frac{dy}{dx}\checkmark = 4 \times \frac{1}{9-3x} \times (-3)\checkmark - 2 \times \frac{1}{x} - \frac{1}{\cos 2x} \times (-\sin 2x)\checkmark$$

$$\frac{dy}{dx} = y\left[-\frac{12}{9-3x} - \frac{2}{x} + \tan 2x\right]\checkmark$$

$$= \frac{(9-3x)^4}{x^2\cos 2x}\left[-\frac{12}{9-3x} - \frac{2}{x} + \tan 2x\right] \quad (4)$$

2.4 $$e^{xy} = \log 4x - e^{5y}$$

$$e^{xy} \times \left[y + x\frac{dy}{dx}\right]\checkmark = \frac{1}{xln10}\checkmark - 5e^{5y}\frac{dy}{dx}\checkmark$$

$$xe^{xy}\frac{dy}{dx} + 5e^{5y}\frac{dy}{dx} = \frac{1}{xln10} - ye^{xy}\checkmark$$

$$\frac{dy}{dx}[xe^{xy} + 5e^{5y}] = \frac{1}{xln10} - ye^{xy}$$

$$\frac{dy}{dx} = \frac{\frac{1}{xln10} - ye^{xy}}{xe^{xy} + 5e^{5y}}\checkmark \quad (5)$$

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QUESTION 3

3.1 3.1.1 $$f(x) = x^3 + x^2 - x + 7$$

$$f'(x) = 3x^2 + 2x - 1 = 0\checkmark$$

$$(x+1)(3x-1) = 0$$

$$x = -1 \ or \ x = \frac{1}{3}$$

$$y = 8 \quad y = \frac{184}{27}$$

Thus, the coordinates of the turning points are $(-1;\ 8)\checkmark$ and $\left(\frac{1}{3}; \frac{184}{27}\right)\checkmark$ (3)

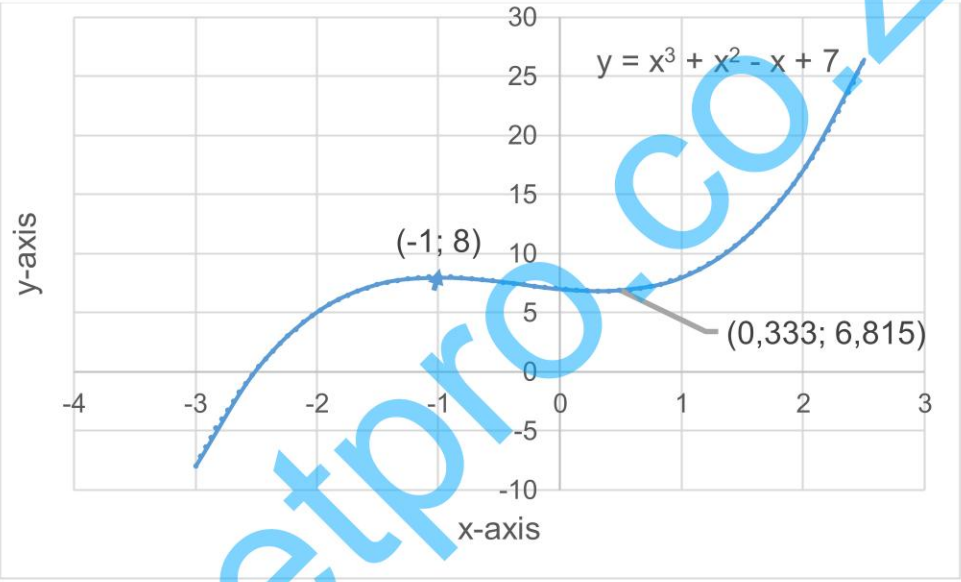
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3.1.2

x	-3	-2	-1	0	1	2	3
y	-8	5	8	7	8	17	40

✓✓= 1 Mark for the every 3 correct answers (2)

3.1.3



1 Mark for the shape
1 Mark for the indication of the turnings point on the graph (2)

3.1.4 $Let\ x_0 = -2{,}7$

$f(-2{,}7) = -2{,}693$

$f'(-2{,}7) = 15{,}47$✓

$x_1 = -2{,}7 - \dfrac{-2{,}693}{15{,}47}$✓

$= -2{,}526$✓ (3)

3.2 $x + y = 18 \ldots\ldots\ldots (1)$✓

$P = x^2y \ldots\ldots\ldots\ldots (2)$✓

$y = 18 - x \ldots\ldots\ldots (3)$

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Substitute (3) into (2)

$P = x^2(18 - x)$

$= 18x^2 - x^3$✓

$P' = 36x - 3x^2 = 0$✓

$3x(12 - x) = 0$

$x = 0 \quad or \quad x = 12$
$y = 18 \qquad y = 6$

For, $x = 0$ and $y = 18$, the product will minimum

Thus, numbers $x = 12$✓ and $y = 6$,✓ will maximise the product. (6)

3.3

$V = \frac{1}{3}\pi r^2 h$

$r = \frac{h}{2}$

$V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$

$V = \frac{1}{12}\pi h^3$✓

$\frac{dV}{dt} = \frac{1}{4}\pi h^2 \frac{dh}{dt}$✓

$15\text{cm}^3/\text{s} = \frac{1}{4}\pi(3\text{cm})^2 \frac{dh}{dt}$✓

$\frac{dh}{dt} = \frac{15}{\frac{9\pi}{4}}$

$= \frac{20}{3\pi}\text{cm/s} \quad \text{or} \quad 2{,}122\text{cm/s}$✓

(4)
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QUESTION 4

4.1 4.1.1

$$\int \frac{\tan x}{\ln(\cos x)}\,dx$$

$let\ u = \ln(\cos x)$
$du = \frac{1}{\cos x}.-\sin x\,dx$✓

$$= -\int \frac{1}{u}\,du$$✓

$$= -\ln u + C$$

$$= -\ln(\ln(\cos x)) + C$$ ✓ (3)

4.1.2 $$\int \cos(ax)\cos(bx)\,dx$$

$$= \int \frac{1}{2}[\cos(a+b)x - \cos(a-b)x]\,dx$$ ✓

$$= -\frac{\sin(a+b)x}{2(a+b)}$$✓ $$-\frac{\sin(a-b)x}{2(a-b)}$$✓ $$+C$$ (3)

4.1.3 $$\int \cos^3 x \sin^2 x\,dx$$

$$= \int \cos x \cos^2 x \sin^2 x\,dx$$ ✓

$$\int \cos x\,(1-\sin^2 x)\sin^2 x\,dx$$ (5)

$u = \sin x$
$du = \cos x\,dx$ ✓

$$= \int (1-u^2)u^2\,du$$ ✓

$$= \int (u^2 - u^4)\,du$$

$$= \frac{1}{3}u^3 - \frac{1}{5}u^5 + C$$

$$= \frac{1}{3}\sin^3 x$$ ✓ $$-\frac{1}{5}\sin^5 x$$ ✓ $$+C$$

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4.1.4 $\int \frac{x^3+4x^2+6}{x+1}dx$

$= \int \left(x^2+3x-3+\frac{7}{x+1}\right)dx$ ✓

$= \frac{1}{3}x^3$✓ $-\frac{3}{2}x^2$✓ $-3x$✓ $+7\ln(x+1)$✓ $+C$ (5)

4.1.5 $\int e^{6x}\sin e^{3x}\,dx$

$let\ u = e^{3x}$	$dv = e^{3x}\sin e^{3x}\,dx$
$du = 3e^{3x}\,dx$	$v = e^{3x}\cos e^{3x}$ ✓

$\int x^5 \ln 3x\,dx = e^{3x} \times -\frac{1}{3}\cos e^{3x} - \int -\frac{1}{3}e^{3x}\cos e^{3x} \times 3e^{3x}\,dx$✓

$= -\frac{1}{3}e^{3x}\cos e^{3x}$ ✓ $+\frac{1}{3}\sin e^{3x}$ ✓ $+C$ (4)

4.2 $\int \frac{2x+3}{(x-2)^2}dx$

$\frac{2x+3}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$

$2x+3 = A(x-2)+B$✓

$let\ x = 2;$	$B = 7$✓
or	
$x = 0;$	$A = 2$✓

$\int \left(\frac{2}{x-2}+\frac{7}{(x-2)^2}\right)dx$

$= 2\ln(x-2)$✓ $-\frac{7}{x+2}$✓ $+C$ (5)

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QUESTION 5

5.1 $\int_0^{\infty} xe^{-2x}\,dx$

$= \lim_{b\to\infty} \int_0^{b} xe^{-2x}\,dx$

$let\ u = x$	$dv = e^{-2x}\,dx$
$du = dx$	$v = -\frac{e^{-2x}}{2}$ ✓

$= \lim_{b\to\infty} \left\{ \left[-x\frac{e^{-2x}}{2} \right]_0^b - \int_0^b -\frac{e^{-2x}}{2}\,dx \right\}$ ✓

$= \lim_{b\to\infty} \left\{ \left(-b\frac{e^{-2b}}{2} - 0 \right) - \left[\frac{e^{-2x}}{4} \right]_0^b \right\}$ ✓

$= 0$ ✓ $- \lim_{b\to\infty} \left(-\frac{e^{-2b}}{4} - \frac{1}{4} \right)$ ✓

$= \frac{1}{4}$ ✓ (6)

5.2 5.2.1 $A = \int_a^b (y_1 - y_2)\,dx$

$= \int_{0,879}^{5,121} [(6x - x^2) - (x^2 - 6x + 9)]\,dx$ ✓

$= \int_{0,879}^{5,121} (6x - x^2 - x^2 + 6x - 9)\,dx$

$= \int_{0,879}^{5,121} (-2x^2 + 12x - 9)dx$

$= \left[-\frac{2}{3}x^3 + 6x^2 - 9x \right]_{0,879}^{5,121}$ ✓

$= 25{,}456\ units^2$ ✓ (3)

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5.2.4

$$V = \pi \int_a^b (y_1{}^2 - y_2{}^2)\, dx$$

$$= \pi \int_{0,879}^{5,121} [(6x - x^2)^2 - (x^2 - 6x + 9)^2]\, dx \checkmark$$

$$= \pi \int_{0,879}^{5,121} (36x^2 - 12x^3 + x^4 - x^4 + 12x^3 - 54x^2 - 108x - 81)dx \checkmark$$

$$= \pi \int_{0,879}^{5,121} (-18x^2 + 108x - 81)dx \checkmark$$

$$= \pi[-6x^3 + 54x^2 - 81x]_{0,879}^{5,121} \checkmark$$

$$= 719{,}747\ units^3 \checkmark$$ (5)

5.3 $I_y = r^2\, dA;\ \ dA = 9dx \checkmark$

$$= x^2 . 9\, dx \checkmark$$

$$= 9\int_0^6 x^2\, dx \checkmark$$

$$= 9\left[\frac{x^3}{3}\right]_0^6 \checkmark$$

$$= 648\ units^4 \checkmark$$ (5)

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QUESTION 6

6.1 $y\frac{dy}{dx} = 3\sqrt{xy^2 + 9x}$, if $y(0) = 4$

$y\frac{dy}{dx} = 3\sqrt{x}\sqrt{y^2 + 9}$

$\frac{y}{\sqrt{y^2 + 9}}dy = 3\sqrt{x}\,dx$

$\sqrt{y^2 + 9} = 2x^{\frac{3}{2}} + C$ ✓

$\sqrt{0^2 + 9} = 2(4)^{\frac{3}{2}} + C$

$C = -13$✓

$\therefore \sqrt{y^2 + 9} = 2x^{\frac{3}{2}} - 13$✓ (3)

6.2 $\frac{d^2y}{dx^2} = 2\left(x - \frac{3}{2}\right)^2 - \frac{1}{2}$, if $\frac{dy}{dx}\Big|_{x=0} = 3$ and $y(1) = 2$

$\frac{d^2y}{dx^2} = 2\left(x - \frac{3}{2}\right)^2 - \frac{1}{2}$

$\frac{d^2y}{dx^2} = 2x^2 - 6x + 4$

$\frac{dy}{dx} = \frac{2}{3}x^3 - 3x^2 + 4x + A$✓

$3 = \frac{2}{3}(1)^3 - 3(1)^2 + 4(1) + A \qquad \therefore A = 3$✓

$\frac{dy}{dx} = \frac{2}{3}x^3 - 3x^2 + 4x + 3$

$y = \frac{1}{6}x^4 - x^3 + 2x^2 + 3x + B$✓

$2 = \frac{1}{6}(1)^4 - (1)^3 + 2(1)^2 - 3(1) + B \qquad \therefore B = -\frac{13}{6}$✓

$y = \frac{1}{6}x^4 - x^3 + 2x^2 + 3x - \frac{13}{6}$✓ (5)

[8]

TOTAL 100

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